

# Yiğit Sarp Avcı

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## Education

**Boğaziçi University - B.Sc. in Computer Engineering**

09/2023 – Present

**Kabataş Boys' High School**

09/2018 – 06/2023

- Graduated with a GPA of 3.78/4.00.
- Actively participated in various extracurricular activities and student societies, including Debate, Model United Nations, Entrepreneurship, and the Coding Club.

## Experience

**Executive Board Member at Radyo Boğaziçi**

09/2023 – Present

- Developed strong skills in team coordination, social media management, budget planning, event organization, communication, and operational decision-making.

**Work and Travel Program Participant**

06/2025 – 09/2025

- Gained valuable international experience through the cultural exchange and work program in the USA which improved my communication, English, teamwork and problem-solving skills.

**Regional Organizational Team Member at ESTIEM**

09/2023 – Present

- Contributed to the planning and coordination of events, team communication, and organizational processes within ESTIEM, an international network connecting Industrial Engineering and Management students across Europe.

## Technical Projects (can be seen at GitHub)

**Performo Enterprise Admin** — Next.js 15+, Prisma, Tailwind CSS, Framer Motion

- High-density administrative intelligence dashboard with real-time operational metrics and server-side synchronized state management.

**AI-Powered FinTech Intelligence Platform** — Next.js, FastAPI, PostgreSQL, Docker

- Full-stack financial dashboard featuring an asynchronous limit-order matching engine and AI-generated insights.

**SchedSim CPU Scheduler** — x86-64 Assembly, GNU Toolchain, Linux Syscalls

- High-performance CPU scheduling simulation engine built with zero libc dependencies using direct Linux syscalls.

**Spieslayer Command Interpreter** — C, Custom Backtracking Lexer

- Robust CLI-based interpreter utilizing an autonomous backtracking lexical parser for complex command resolution.

**Aura Functional Interpreter** — Python, Functional Paradigms, Dual Scoping Engine

- Advanced functional language interpreter supporting first-class closures and toggleable lexical/dynamic scoping engines.

**MatrixNet** — Java, Tarjan's SCC Algorithm, Graph Theory

- Network graph analysis tool implementing Tarjan's SCC and Articulation Point detection for infrastructure vulnerability detection.

**GigMatch-Pro** — Java, Memory-Optimized Heaps

- Scalable task-matching marketplace engine utilizing optimized dynamic Max-Heaps for high-efficiency priority allocation.

**Nightpass Survival Game** — Java, Rule Engine Architecture

- Algorithmic rule engine governing strategic resource economics and state-machine logic in a terminal-based survival game.

Volunteer Experience

Education Mentor at TOÇEV09/2019 – 03/2020

- Provided volunteer tutoring and mentoring for children from disadvantaged backgrounds.

Volunteer at Kabataş Boys’ High School Community Involvement Projects Club09/2018 – 06/2023

- Carried out various volunteer projects aimed at helping and supporting people in need through community involvement initiatives.

Volunteering Division Member at Boğaziçi University Industrial Engineering Club09/2023 – 06/2024

- Supported the organization and execution of volunteer initiatives and community-oriented projects within the club.

Education Mentor at Derece Kampüsü

- Mentored high school students for the university entrance exams, providing academic guidance and exam strategy support.

Technical Skills

- **Low-Level & Systems:** x86-64 Assembly, C, Linux Syscalls, OS Internals (Scheduling, Pthreads, Memory Mgmt)
- **Enterprise Full-Stack:** Next.js 15+, TypeScript/React, FastAPI, ORM (Prisma, SQLAlchemy), PostgreSQL/SQLite, Tailwind & Framer Motion
- **Algorithms & Theory:** Graph Theory (Tarjan's SCC), Advanced Data Structures (Heaps), Complexity Analysis (Big O), Recursive Optimization
- **Paradigms & Models:** Functional (Scheme), Logic (Prolog), OOP (Java/C++), Interpreter Design, Finite State Machines
- **AI & Agentic Systems:** Advanced Agentic Coding, AI-Assisted Architecture, LLM Engineering (OpenAI API, Prompting)
- **Engineering Ecosystem:** Linux/Unix, Git/GitHub, Docker, CI/CD, Alembic, Pytest, Makefile, LaTeX

Languages

- Turkish (native)
- English (fluent)
- German (elementary)

Extracurricular Activities

- DJ, Radyo Boğaziçi
- Student-Athlete, Boğaziçi University Sailing Team
- Student-Athlete, Boğaziçi University Ski Team
- Windsurf Athlete, Competing Individually